

Practice Midterm Exam 1

Kühnel, Chapters 2–3

75 minutes

Given (first) name: _____

Family (last) name: _____

BU ID: _____

Instructions

- Allow 75 minutes for timed practice. **Choose any four of the five problems.** Each is worth 20 points, for a total of 80 points. Indicate the omitted problem below; only your four selected problems will be graded.
- Show your calculations and justify your claims. State clearly any theorem you use. You may use results from Chapters 2–3 unless the problem asks you to prove them.
- All parts of a problem are listed together. Use the space below the questions and the blank work pages between problems. Clearly label any work continued elsewhere.
- Give exact answers; no electronic assistance is needed.
- Dots denote derivatives in an arbitrary parameter t ; primes denote derivatives in arclength s .
- For an oriented surface, use the shape operator $S = -d\nu$, Gaussian curvature $K = \det S$, and mean curvature $H = \frac{1}{2} \operatorname{tr} S$.

Problem omitted (1–5): _____

Problem	1	2	3	4	5	Total
Points possible	20	20	20	20	20	80
Score						

This booklet contains 10 pages: the cover, five problem pages,
and one blank work page between each pair of problems.

Problem 1.**[20 points]**

Consider the curve

$$c(t) = (2 \cos t, 2 \sin t, t), \quad t \in \mathbb{R}.$$

- (a) Compute the speed, the arclength function with $s(0) = 0$, and the length of the segment $0 \leq t \leq 2\pi$. **[5 points]**
- (b) Compute the curvature κ and torsion τ . You may use the standard formulas for an arbitrary regular parameter. **[9 points]**
- (c) Let $\tilde{c} : \mathbb{R} \rightarrow \mathbb{R}^3$ be a unit-speed Frenet curve with the same curvature and torsion functions as the arclength parametrization of c . Explain precisely what the fundamental theorem of curves implies about the relation between these two curves. **[6 points]**

Problem 2.**[20 points]**

Let $c : I \rightarrow \mathbb{R}^3$ be a smooth unit-speed Frenet curve on an interval, with $\kappa > 0$ and $\tau \neq 0$ everywhere. Write $\rho = 1/\kappa$. You may use the Frenet equations

$$T' = \kappa N, \quad N' = -\kappa T + \tau B, \quad B' = -\tau N.$$

- (a) Suppose c lies on a sphere with fixed center m . By differentiating $\|c - m\|^2$, show that

$$m = c + \rho N + \frac{\rho'}{\tau} B.$$

[10 points]

- (b) Prove that c lies on a sphere if and only if

$$\left(\frac{\rho'}{\tau}\right)' + \rho\tau = 0.$$

In the converse direction, explain why both the center and the distance from c to that center are constant. Do not quote the spherical-curve criterion. **[10 points]**

Problem 3.**[20 points]**

Let

$$f(u, v) = (u, v, uv), \quad (u, v) \in \mathbb{R}^2,$$

and let $M = f(\mathbb{R}^2)$.

- (a) Show that f is an embedding, and find the affine tangent plane to M at $p = f(1, 1) = (1, 1, 1)$. **[6 points]**
- (b) Compute E, F, G and show that the metric is positive definite. **[6 points]**
- (c) Compute the angle between f_u and f_v at $(1, 1)$. **[3 points]**
- (d) Set $\phi(r, s) = (r + s, r - s)$ and $\tilde{f} = f \circ \phi$. Use the coordinate-change formula to compute the matrix of the first fundamental form of $\tilde{f}$ at $(r, s) = (1, 0)$. Are its coordinate tangent vectors orthogonal there? **[5 points]**

Problem 4.**[20 points]**

Consider the helicoid

$$f(u, v) = (u \cos v, u \sin v, v), \quad (u, v) \in \mathbb{R}^2.$$

Use the unit normal $\nu = (f_u \times f_v) / \|f_u \times f_v\|$.

(a) Show that f is an immersion. Compute E, F, G and ν .

[7 points]

(b) Compute the coefficients of the second fundamental form:

$$e = \langle f_{uu}, \nu \rangle, \quad b = \langle f_{uv}, \nu \rangle, \quad g = \langle f_{vv}, \nu \rangle.$$

[5 points]

(c) Compute the Gaussian curvature and mean curvature at every point. You may use

$$K = \frac{eg - b^2}{EG - F^2}, \quad H = \frac{eG - 2bF + gE}{2(EG - F^2)}.$$

[5 points]

(d) Explain why the helicoid is both a ruled surface and a minimal surface. Identify which coordinate curves are the straight rulings.

[3 points]

Problem 5.**[20 points]**

At a point p of an oriented surface, let e_1, e_2 be an orthonormal basis of principal directions, with principal curvatures k_1, k_2 . The normal curvature in a unit direction X is $k_n(X) = \langle SX, X \rangle$. A nonzero direction with zero normal curvature is called *asymptotic*.

- (a) For $X = \cos \theta e_1 + \sin \theta e_2$, derive

$$k_n(X) = k_1 \cos^2 \theta + k_2 \sin^2 \theta.$$

[5 points]

- (b) Assume $K(p) < 0$, and label the principal curvatures so that $k_1 > 0 > k_2$. Show that there are exactly two distinct unoriented asymptotic tangent lines at p , and prove that they are perpendicular if and only if $H(p) = 0$.

[10 points]

- (c) After reversing the unit normal, how do the principal curvatures, K , H , and the asymptotic tangent lines change? Justify your answers.

[5 points]

End of exam.